

The planar Tingley problem is answered a literature-status note for arXiv:2005.04794

Literature identification packet

13 August 2026

Status: literature already answered. The introductory two-dimensional open statement in arXiv:2005.04794 was explicitly resolved by Taras Banakh, arXiv:2103.09268, Theorem 1.3.

Original statement

Cueto-Avellaneda and Peralta state in the first paragraph of the introduction to *Can one identify two unital JB^* -algebras by the metric spaces determined by their sets of unitaries?* (arXiv:2005.04794, page 1) that Tingley’s problem “remains open even for two dimensional spaces.”

In precise terms, the planar problem asks: if X and Y are two-dimensional Banach spaces and

$$f : S_X \longrightarrow S_Y$$

is a surjective isometry between their unit spheres, must f extend to a surjective real-linear isometry $X \rightarrow Y$? Equivalently, does every two-dimensional Banach space have the Mazur–Ulam property?

Decisive later answer

Banakh’s *Every 2-dimensional Banach space has the Mazur–Ulam property* (arXiv:2103.09268) restates Tingley’s question as Problem 1.1. Its main result is:

Theorem 1.3. Every 2-dimensional Banach space has the Mazur–Ulam property.

The abstract explicitly says that every isometry between the unit spheres of two-dimensional Banach spaces extends to a linear isometry and that this resolves Tingley’s problem in dimension two. Hence it is an exact affirmative answer to the source’s introductory open statement.

Provenance and scope

This packet records a direct later-literature answer, not an independently derived result. The supporting author explicitly identifies the theorem as a resolution of the same problem. The identification is supported by the exact problem formulation, theorem label, and matching dimensional scope.

The source paper’s title question, concerning isometries of unitary sets of unital JB^* -algebras, is answered inside arXiv:2005.04794 itself and is not the signal recorded here. Banakh’s theorem resolves only the two-dimensional case; the unrestricted Tingley problem is not claimed to be solved by this packet.

References

M. Cueto-Avellaneda and A. M. Peralta, *Can one identify two unital JB^* -algebras by the metric spaces determined by their sets of unitaries?*, arXiv:2005.04794. <https://arxiv.org/abs/2005.04794>.

T. Banakh, *Every 2-dimensional Banach space has the Mazur–Ulam property*, arXiv:2103.09268, Theorem 1.3. <https://arxiv.org/abs/2103.09268>.